SEQUEL → English

SQL : Structured Query Language :

→ to interact with database.

→ special software data management.

Oracle

MySQL

DBMS : Data base management system

→ secure

→ concurrent transactions.

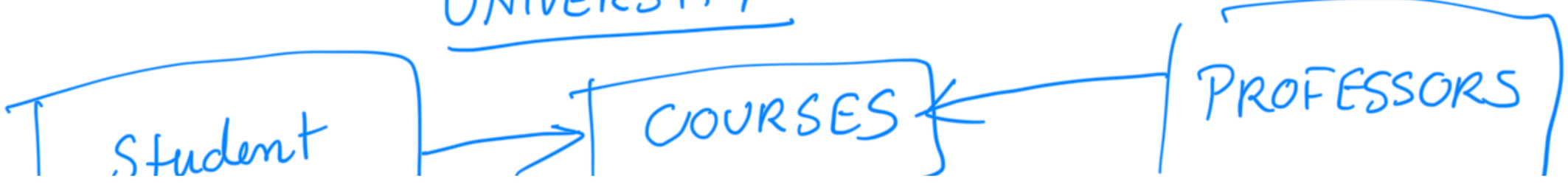
→ errors handling.

SQL ✓✓✓

clause

- SELECT → column selection df[['col', 'col']]
- FROM → table df
- WHERE → filter - condition [df['country'] == 'Japan']
- GROUP BY → group categories.
- ORDER BY → sort sort_values( )
- LIMIT → head( )

UNIVERSITY



is a collection of
inter-related tables.

MySQL : setup.

Bigquery : GCP

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① conditions : multiple

AND

OR

cond 1	cond 2	AND	OR
TRUE	TRUE	TRUE	TRUE
TRUE	FALSE	FALSE	TRUE
FALSE	TRUE	FALSE	TRUE
FALSE	FALSE	FALSE	FALSE

age > 35

Group By and HAVING

```
SELECT COUNTRY, COUNT(CUSTOMER_ID)
```

FROM CUSTOMERS

~~WHERE COUNT(CUSTOMER_ID) > 5~~ ✗

GROUP BY COUNTRY

HAVING COUNT(CUSTOMER_ID) > 5 ✓